Physical Integration Guide for Tiliter Vision 4.0 Camera

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1 Background

The Tiliter Vision system uses a machine learning model that recognises products. This model is trained to be generalised allowing highly accurate recognition in a wide variety of ambient lightning brightness, color temperature, backgrounds, product rotations/positions and camera perspectives.

Certain camera angles are widely used and have more real-world data available to validate recognition accuracy performance.

2 Purpose

The purpose of this document is to describe the preferred camera positions and angles, and provide examples of existing mounting options currently available. The contents shall enable the reader to make educated decisions on how to mount the Tiliter Vision Camera for optimal machine vision performance.

3 Scope

Guidance on preferred physical integration of the TV4 Camera.

4 Audience

This document is intended for internal and external distribution. The key audience are Engineers tasked with the integration of the Tiliter Vision Camera to existing physical infrastructure or in lab environments where the product performance is being setup for proving.

5 Preferred Camera Position

Tiliter prefers a top-down camera position.

When the camera is positioned higher up with a larger angle from horizontal it is less likely to incidentally capture the environment or sensitive information of the shopper. Using top-down position like this also results in improved image clarity and prediction performance because depth-of-field is consistent and article sizes are less distorted. Lower camera positions can impact performance, and also require privacy controls, such as intentional background redaction.

Due to the physical constraints of the existing solutions, a lower camera position may be the only feasible solution and as such these lower viewing angles are also acceptable.

6 Preferred Camera Field of View

To ensure the best results, the camera FOV should be centered to the typical placement of the article and the capture of distant non-article features minimized, refer to Figure 1. The camera FOV shall also be held level to the ground. The Camera FOV is 60° x 30° and can be viewed via the stream on the system dashboard. To view the dashboard, enable internet sharing to the ethernet port which the camera processor is connected and type the following in the browser “<hostname>.local:8000/dashboard”. An example of the hostname is “TV040303AU1000000303”.

For more information on system setup refer to the Tiliter Vision HTTP Interface Design Description document.

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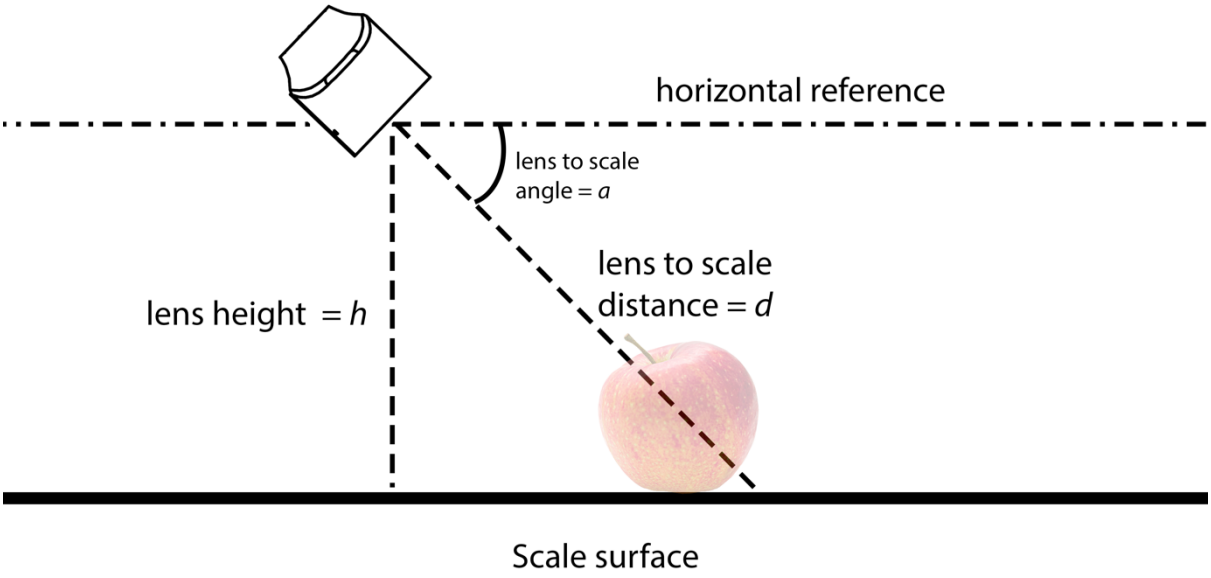


Figure 1: Reference diagram with camera position measurements

Notably the field-of view can be cropped with software modifications if required, contact respective Tiliter persons for more information.

7 Field-Proven Camera Positions

There are a number of field-proven camera positions available. These positions align with machine vision training images in Tiliter's article database. In theory, the field-proven positions provide the best performance for identifying products second to near top-down perspectives.

Reference mounting positions are found below:

Name	**Lens height *h* (mm)**	**Lens to scale angle from horizontal *a* (deg)**	**Resultant lens to scale distance *d* (mm)**
Preferred	260	60	300
Dev Kit	180	40	280
Low Profile 1	125	37	208
Low Profile 2	96	28	204

Table 1: Camera Position Measurements

Refer Figure 1 and Figure 2 to interpret the camera mounting position measurements.

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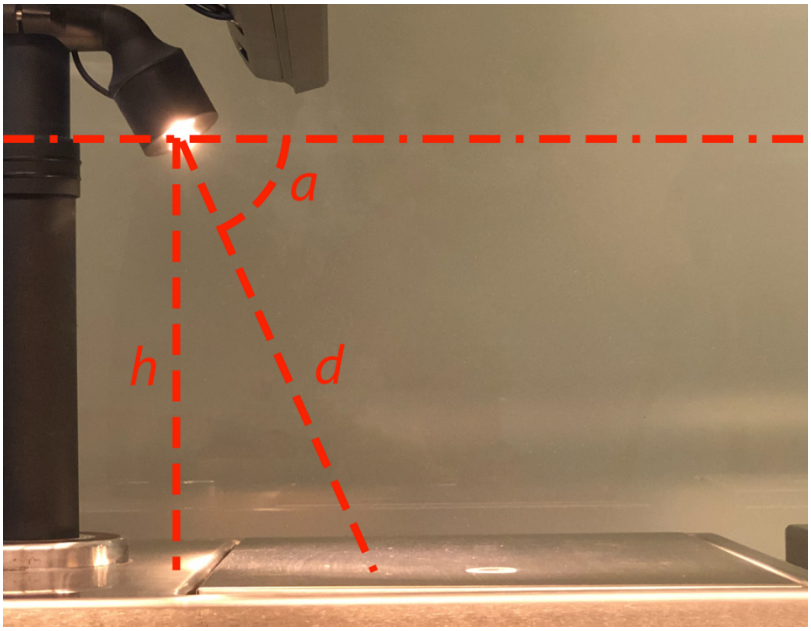


Figure 2: Real world example showing the same measurements

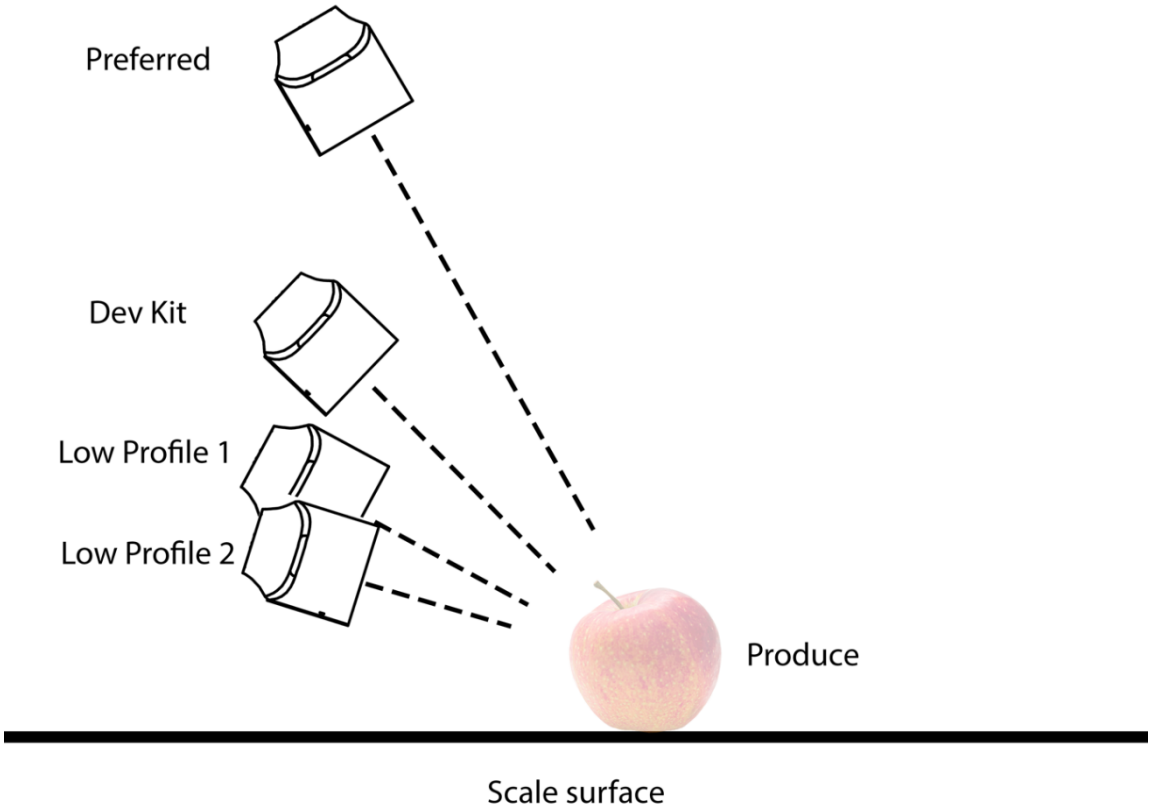
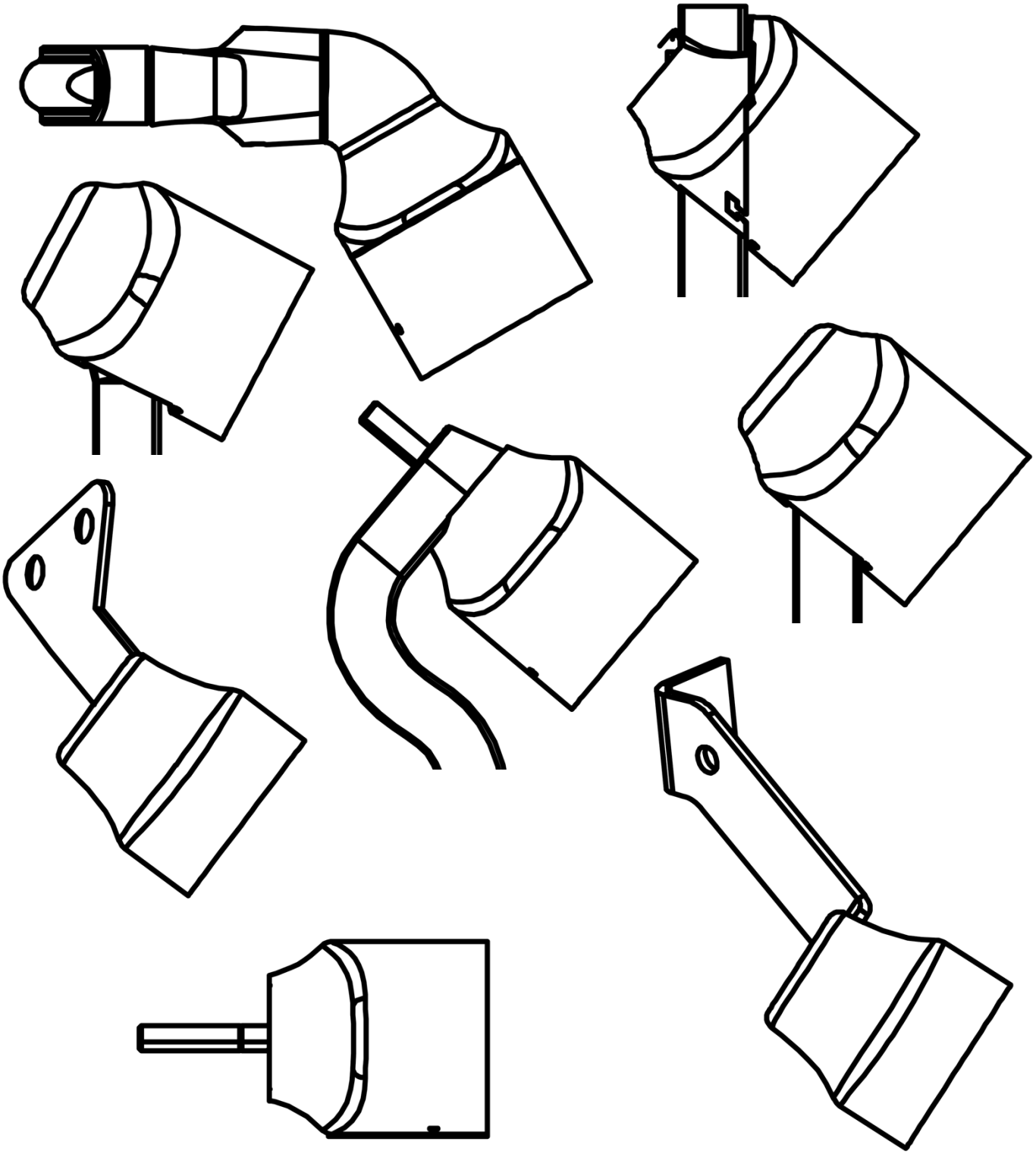


Figure 3: Visualization of the field-tested camera positions

8 Existing Camera Mounting Solutions

Tiliter have a wide range of existing camera mounts that can be used to connect to common POS checkouts or scales. Existing infrastructure such as the mount for the Datalogic TDR arm, or a display mounting pole can be used.

Below are some examples of what have been designed inhouse:



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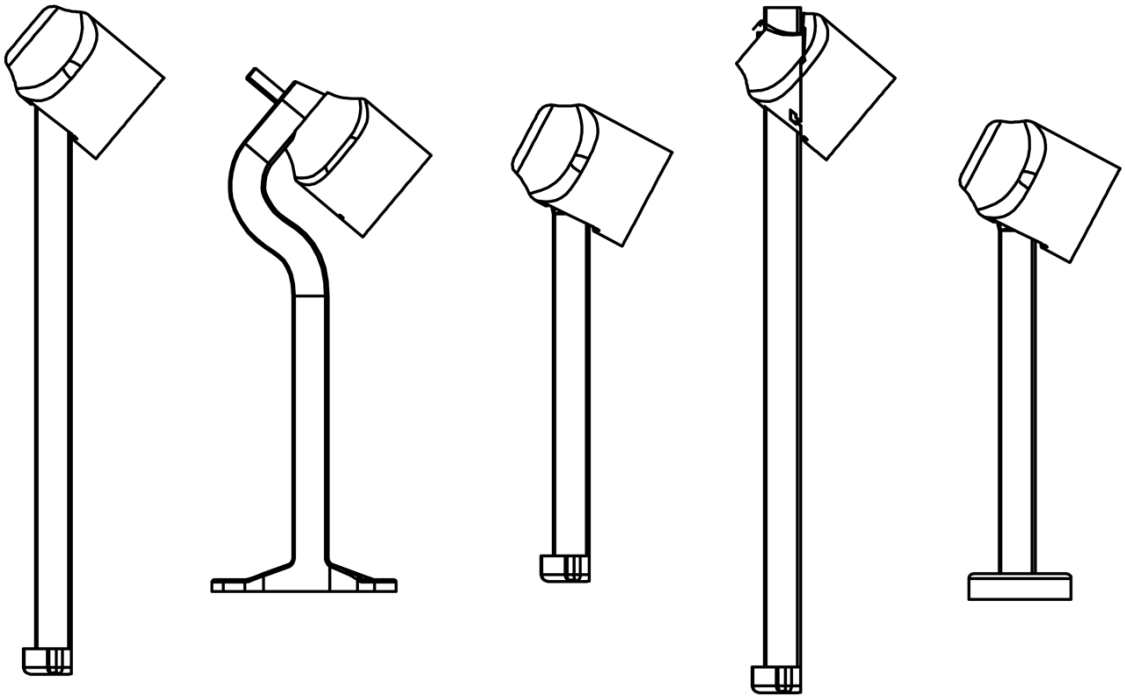


Figure 4: Examples of Existing Camera Mounting Solution

9 Customer Designed Mounting Solutions

Tilter offers a versatile camera mounting design which is intended to be easily integrated by customer engineers where an existing solution does not yet existing. The camera has an M6 bolt at the back with a ball-socket joint that provides up-to ~15° of pivoting motion and 360° of rotational motion. The bolt thread extends out by ~34mm. The intention is that the customer would design a bracket which the M6 bolt can be fastened to with M6 nut.

Notably the versatile mounting solution is subject to accidental misalignment from a desired perspective eg. nudge from shopper. Therefore, Tiliter recommends the degrees of motion are locked-in once the optimum FOV / Perspective has been dialed-in. This can be achieved with a grub screw and surface that sits flush with the back of the camera. Prototyping with sheet metal and additive fabrication is recommended.

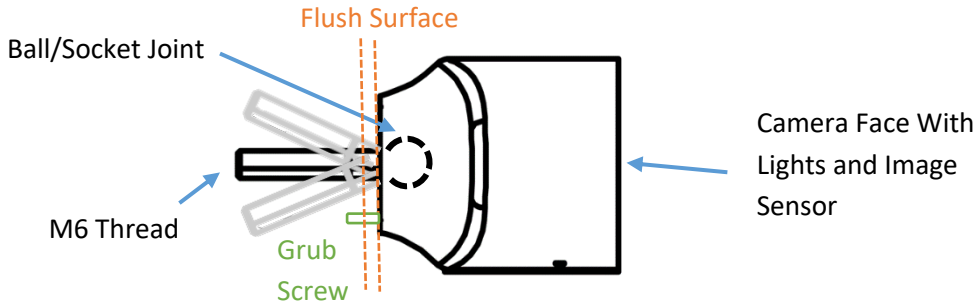


Figure 5: Versatile Camera Mounting Solution Side View

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10 Custom Physical Integration as a Service

Tiliter offers a service for designing and manufacturing of integration solutions that have not yet been developed. This service is available to customers seeking to integrate into their solutions, but they do not have the resources or expertise in-place to execute such a task.

This process is executed in five parts. Responsibility assigned to each of these parts can be assigned where required to Tiliter.

- Scouting – Physically obtaining dimensions / constraints of existing system
- Design – Design the product in CAD
- Prototyping – Manufacture the first solution with rapid prototyping tech
- Validation – Check perspective / durability is suitable in retail – confirm with customer.
- Manufacturing – Manufacture small batch

11 Camera Focus

The Tiliter Vision 4.0 Cameras have a manually adjustable focus, Tiliter can factory set these to the required focal distance, which is calculated as “d” (refer to Figure 1) minus 4cm.

It is likely the focus will need to be adjusted by the customer if they use custom designed integration. The optimal focal distance can be manually adjusted in a special user interface mode that shows the clarity of the image and a special tool that fits onto the lens.

Rotating the tool adjusts the focal length which results in more and less red particles on the stream when viewed via the special user interface. The red particles on the screen indicate that sharp edge has been identified which is a measure of focus. Higher red particle counts indicate a sharper image. There is also a number that corresponds to the number of red particles. The focus should be adjusted to find the highest attainable number.

The special user interface mode can be accessed at “<hostname>.local:8000/dashboard” and then toggling the focus info switch.

The subject used for judging focus shall be approximately 4cm from platter height if viewing top-down perspective. The subject shall be high contrast with many small / sharp details. An example subject for judging focus is shown in Figure 7.

Figure 6: Focus Tool

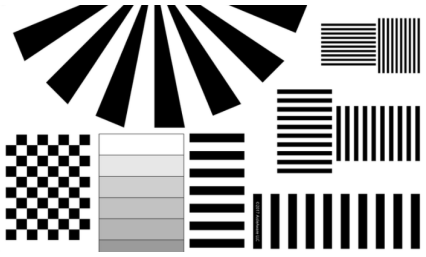


Figure 7: Example Focus Subject

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12 Preferred Placing Surface

To ensure optimal performance the article placing surface should not take any attention from the article. This can be achieved by the following surface parameters:

- Consistent colour and finish
- Dull finish
- Non-reflective
- Avoiding geometric shapes and patterns

Notably, brushed stainless steel placing surfaces have been proven successful in-field and some existing models have been trained specifically for more complex placing surfaces. Tiliter can train predication models when surface parameters do not meet the requirements stated above. Contact respective Tiliter Representative if this is required.

13 Preferred Lighting

Tiliter Vision 4.0 Camera's have integrated lights that provide flood lighting during the prediction, these lights can be set at 100% duty (always on) if required. The intention of these lights is to overpower the environmental lighting to provide consistent; colour, brightness and white balance. Field testing has shown that strong environmental lighting which is being projected from behind the camera and onto the test subjects results in decreased prediction performance. As such it is recommended strong environmental lighting from behind the system is minimized.

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14 Revision History

Revision	Date	Author	Approver	Changes
1	03/12/2020	Ben Weir	Detlev Magura	Initial release

15 References

Reference	Description
Tiliter Vision HTTP Interface Design Description	Document describing the Tiliter Vision web API

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